

General_TRW_permutation_channels_acw_0_block

July 20, 2025

```
[38]: ##modules
      %%matplotlib widget
      %%matplotlib inline
      #
      %matplotlib qt
      import mne
      import numpy as np
      import matplotlib
      # Establecer un backend interactivo, como 'Qt5Agg', 'GTK3Agg', etc.
      # Esto depende de los backends disponibles en tu sistema.

      import matplotlib.pyplot as plt

      #matplotlib.use('TkAgg') # Asegúrate de que este backend está instalado.

      import pandas as pd
      import os
      import sys

      from mne.preprocessing import ICA, corrmap, create_ecg_epochs, create_eog_epochs

      from os.path import join as pathjoin
      from time import time

      from pathlib import Path

      from autoreject import AutoReject

      # aplicar la acf EN epochs

      from statsmodels.tsa.stattools import acf
      import numpy as np

      import pandas as pd
```

```

from scipy.stats import permutation_test
sys.path.append(r"G:\apuntes_mne\code_MOUS\scripts_functions_created")

# Ahora importa la función
from print5 import print5

```

```

[39]: try:
        # Si se ejecuta como SCRIPT .py: usar __file__
        sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), "..
        ↪")))
    except NameError:
        # Si se ejecuta como NOTEBOOK Jupyter: usar path relativo
        sys.path.append("..") # sube un nivel desde la carpeta actual del notebook

# --- Configuración dinámica de rutas ---
from get_paths_MOUS import get_paths_MOUS

# Parámetros editables
disco = "g"
modality = "visual"
layer_script = "block"
subj = "sub-V1001"
#subj = sys.argv[1] ## name of participant list

# Generar variables automáticamente
path_dict = get_paths_MOUS(disco=disco, modality=modality,
    ↪layer_script=layer_script, subj=subj)
globals().update(path_dict)

# Mostrar todos los paths generados
print("\n Rutas generadas:")
for k, v in path_dict.items():
    print(f"{k:<20} → {v}")

```

```

Rutas generadas:
datadir           → g:\MOUS_204\MOUS_visual
general_datadir   → g:\MOUS_204
output_preproc    → g:\MOUS_204\MOUS_visual\output_preproc
mri_dir           → g:\MOUS_204\sub-V1001\anat
meg_dir           → g:\MOUS_204\sub-V1001\meg
preproc_path      → g:\MOUS_204\MOUS_visual\output_preproc\preproc_block
channels_structure_path → g:\MOUS_204\channels_structure
epochs_path       →
g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\epochs_block
ICA_path          →

```

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g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\ICA_block
epochs_clean_path      →
g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\epochs_clean_block
evoked_path            →
g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block
source_path            → g:\MOUS_204\MOUS_visual\output_source\source_block
raw_hsp_path           →
g:\MOUS_204\MOUS_visual\output_source\source_block\raw_hsp
fwd_path               → g:\MOUS_204\MOUS_visual\output_source\source_block\fwd
inverse_path          →
g:\MOUS_204\MOUS_visual\output_source\source_block\inverse
output_analysis        → g:\MOUS_204\MOUS_visual\output_analysis
analysis_path          → g:\MOUS_204\MOUS_visual\output_analysis\analysis_block
ACW_path              →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\acw_block
PLE_path              →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\PLE_block
ISC_path              →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\ISC_block
ISC_block_path         →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\ISC_block

```

1 Tables and paths

1.0.1 Paths

ACW_path PLE_path

1.0.2 Tables and metrics

Block ACW f"acw_0_subjects_all_{layer_script}.pickle"

f"acw_50_subjects_all_{layer_script}.pickle"

f"autocorrelation_subjects_all_{layer_script}.pickle"

variables = acw_50_elect_all_epoch_all // 'intercept_y_elect_all_epoch_all',
'slope_elect_all_epoch_all'

PLE f"table_PLE_slope_intercept_subjects_all_{layer_script}.pickle"

f"table_PLE_subjects_all_{layer_script}.pickle"

dynamic f"dynamic_autocorrelation_results_subjects_all_{layer_script}.pickle"

f"dynamic_autocorrelation_subjects_all_{layer_script}.pickle"

variables = acw_50_slope_elect_all_epoch_all // acw_50_std_elect_all_epoch_all

[40]: *## Variables of script*

```

##path=analysis_path
path = ACW_path
name_table = f"acw_0_subjects_all_{layer_script}.pickle"
table_df = pd.read_pickle(f"{path}\\{name_table}")

#METRIC NAMES
metric='acw_0_elect_all_epoch_all'

if '_elect_all_epoch_all' in metric:
    print("yes")
    metric_name = metric.replace('_elect_all_epoch_all', "")
else:
    metric_name = metric

##CONDITION NAMES

condition_1_name="zinnen"
condition_2_name="woorden"

condition_names = [condition_1_name, condition_2_name]

#TYPE OF DIFFERENCES
difference = "inverse" # "normal" or "inverse"

## this will be used in permutation differences
if difference == "normal":
    alternative = "greater"
    # this is for cluster permutation test
    tail=1
    threshold_direction =1
else:
    alternative="less"
    tail=-1
    threshold_direction ==-1

channels = pd.read_csv(channels_structure_path / f"channels_mag_{modality}.csv")
channels_mag=channels[channels[f"canal_efectivo_{modality}"].
    notna()][f"canal_efectivo_{modality}"]

channels_mag=channels_mag.tolist()
print5(channels_mag)

```

```

indice_channels_efectivos = channels[channels[f"canal_efectivo_{modality}"].
↳notna()]["indice"]
indice_channels_efectivos=indice_channels_efectivos.tolist()
# del channels

##valores de las columnas
condition = table_df["Condition"].unique()
print5("Condiciones en los datos:", condition)

subjects = table_df["Subject"].unique()
print5("Sujetos en los datos:", subjects)

elect_all= table_df["Elect"].unique()
print5("sensores en los datos:", elect_all)

epochs_all= table_df["Epoch"].unique()
print5("Epochs en los datos:", epochs_all)
## get info

#read epochs to build evoked
epochs_zinnen=mne.read_epochs(epochs_clean_path /
↳f"{subj}_epochs_zinnen_{layer_script}-epo.fif")
len(epochs_zinnen.pick("mag", exclude="bads").ch_names)

evokeds_zinnen=epochs_zinnen.average()
## evokeds_zinnen es una lista

##cojo el primer elemento, solo tengo una lista
evoked_zinnen=evokeds_zinnen

info=evoked_zinnen.info
del epochs_zinnen
del evokeds_zinnen
del evoked_zinnen

```

yes

```

['MLC11-4304', 'MLC12-4304', 'MLC13-4304', 'MLC14-4304', 'MLC15-4304',
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 'MZ002-4304', 'MZ003-4304', 'MZO01-4304']

Condiciones en los datos: ['zinnen' 'woorden']

Sujetos en los datos: ['sub-V1001' 'sub-V1002' 'sub-V1005' 'sub-V1007'
 'sub-V1008' 'sub-V1009'

'sub-V1010' 'sub-V1011' 'sub-V1012' 'sub-V1013' 'sub-V1016' 'sub-V1019'

'sub-V1020' 'sub-V1022' 'sub-V1024' 'sub-V1029' 'sub-V1030' 'sub-V1031']
sensores en los datos: ['MLC11-4304' 'MLC12-4304' 'MLC13-4304' 'MLC14-4304'
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'MLC16-4304' 'MLC17-4304' 'MLC21-4304' 'MLC22-4304' 'MLC23-4304'
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'MRT13-4304' 'MRT14-4304' 'MRT15-4304' 'MRT16-4304' 'MRT21-4304'
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'MZ002-4304' 'MZ003-4304' 'MZP01-4304']
```

Epochs en los datos: [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
19 20 21 22 23]

Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\epochs_clean_block\
sub-V1001_epochs_zinnen_block-epo.fif ...

Found the data of interest:

t = 0.00 ... 60000.00 ms

0 CTF compensation matrices available

Not setting metadata

24 matching events found

No baseline correction applied

0 projection items activated

```
[41]: #lectura de los diccionarios ISC_zinnen y ISC_woorden
dict_woorden_block = pd.read_pickle(ISC_block_path /f"dict_woorden_block.pkl")
dict_zinnen_block = pd.read_pickle(ISC_block_path /f"dict_zinnen_block.pkl")
```

```
[42]: ##channels significant adjusted in each condition
numbers_channels_woorden=[]
↳dict_woorden_block["significant_channels_adjusted_woorden"]
numbers_channels_zinnen=[]
↳dict_zinnen_block["significant_channels_adjusted_zinnen"]
print("numbers_channels_woorden", numbers_channels_woorden,↳
↳"numbers_channels_zinnen", numbers_channels_zinnen)
print("len(numbers_channels_woorden)",len(numbers_channels_woorden), ↳
↳"len(numbers_channels_zinnen)",len(numbers_channels_zinnen))

names_channels_woorden=[channels_mag[i] for i in numbers_channels_woorden]
names_channels_zinnen=[channels_mag[i] for i in numbers_channels_zinnen]

print("names_channels_woorden", names_channels_woorden)
print("names_channels_zinnen", names_channels_zinnen)

#establecimiento de canales palabras, canales frases y canales mixtos
numbers_channels_only_woorden = list(
    set(numbers_channels_woorden) - set(numbers_channels_zinnen)
)

numbers_channels_only_zinnen = list(
```



```

    set(numbers_channels_zinnen) - set(numbers_channels_woorden)
)

```

```

number_channels_intersection = list(
    set(numbers_channels_woorden) & set(numbers_channels_zinnen)
)

```

```

print(f"len(significant_channels_only_woorden):_
↳{len(numbers_channels_only_woorden)},\
    len(significant_channels_only_zinnen):_
↳{len(numbers_channels_only_zinnen)},\
    len(significant_channels_intersection)_
↳{len(number_channels_intersection)}")

```

```

#create list of names of channels

```

```

names_channels_only_woorden = [channels_mag[i] for i in_
↳numbers_channels_only_woorden]
names_channels_only_zinnen = [channels_mag[i] for i in_
↳numbers_channels_only_zinnen]
names_channels_intersection = [channels_mag[i] for i in_
↳number_channels_intersection]

```

```

numbers_channels_woorden [ 1  14  23  46  52  56  58  59  60  63  67  71  81
82 83 84 86 87
    88 89 90 92 93 94 112 118 122 133 140 141 144 145 150 180 184 192
    201 210 213 215 220 221 224 237 240 252 254 256 269 270]
numbers_channels_zinnen [ 10  13  29  31  32  36  37  40  41  45  46  56  60  72
79 82 83 84
    86 87 88 89 94 105 106 112 117 118 123 135 136 140 141 143 144 145
    155 167 173 199 200 211 212 215 216 217 219 223 228 229 230 234 235 242
    243 244 250 251 256]
len(numbers_channels_woorden) 50 len(numbers_channels_zinnen) 59
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len(significant_channels_only_woorden): 31,
len(significant_channels_only_zinnen): 40,
len(significant_channels_intersection) 19
```

1.1 Posibles comparaciones estadísticas

```
len(significant_channels_only_woorden): 25, len(significant_channels_only_zinnen): 92,
len(significant_channels_intersection) 88
```

Primer problema, el numero de canales zinnen es claramente superior al de woorden, big problem, aunque la interseccion es alta, tal como esperaríamos. Seguramente esto sea porque hay ruido

- comparación ACW-50 en zinnen entre estos canales zinnen y canales woorden + comapracion en woorden de lo mismo
- comparación entre solo zinnen con intersection en woorden y en zinnen (aunque ojo, esto es básicamente aceptar que los canales solo Zinnen son ruido)
- comapración en el promedio general de acw en ambas condiciones de los canalaes zinnen vs promedio general de canales woorden
- comparación de la intersección vs canales no significativos: no sé muy bien como interpretar esto
-

```
[43]: #filtras por sujeto
channels_type=["ch_only_woorden", "ch_only_zinnen", "ch_intersection"]

dict_metric_condition_subj_epoch_mean_ch_type = {}
dict_metric_condition_subj_epoch_mean_ch_type_mean = {}

for cond in condition_names:
    for ch_type in channels_type:
        # for subj in subjects:
            #creacion de lista de valores de metric para cada sujeto

            #eliges la condicion y el tipo de canal y te quedas con los valores
            if ch_type == "ch_only_woorden":
                elects = names_channels_only_woorden
            elif ch_type == "ch_only_zinnen":
```

```

        elects = names_channels_only_zinnen
    elif ch_type == "ch_intersection":
        elects = names_channels_intersection

    array_metric_condition_subj_epoch_mean_ch_type = (
        table_df
        .query("Elect in @elects")
        .query("Condition == @cond")
        .groupby(["Subject", "Elect"])[metric]
        .mean().reset_index()
        .pivot(index="Subject", columns="Elect", values=metric)
        .to_numpy()
    )

    array_metric_condition_subj_epoch_mean_ch_type_mean = (
        table_df
        .query("Elect in @elects")
        .query("Condition == @cond")
        .groupby("Subject")[metric]
        .mean()
        .values
    )

    dict_metric_condition_subj_epoch_mean_ch_type[f"X_{cond}_{ch_type}"] =
    ↪array_metric_condition_subj_epoch_mean_ch_type

    ↪
    ↪dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{cond}_{ch_type}"] =
    ↪array_metric_condition_subj_epoch_mean_ch_type_mean

```

```
[44]: dict_metric_condition_subj_epoch_mean_ch_type
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```
0.16608187, 0.09532695,
    0.26176274, 0.06842941, 0.27606407, 0.23685008, 0.19395881,
    0.0862201 , 0.30498538, 0.2992807 , 0.27014354, 0.06065789,
    0.28907704, 0.14104466, 0.17280702]}}
```

2 1. Comparaciones ch_Zinnen only vs ch_intersection

These are gonna be comparisons between channels zinnen vs intersection in each condition. E.g. in zinnen condition, we compare acw in

```
[46]: from scipy.stats import permutation_test

##dependent condition
def paired_statistic(diff, _):
    return np.mean(diff)

# Independent condition
def diff_means(x, y):
    return np.mean(x) - np.mean(y)

experimental_condition=["zinnen", "woorden"]
type_channel=["intersection", "only_woorden"]
```

2.1 1.1 Comparisons in each experimental condition

2.1.1 1.1.1 If we assume dependency

```
[47]: print("DEPENDENT analysis in {metric_name}")
for exp in experimental_condition:
    print(f"in {exp} condition")
    for ch in type_channel:
        print(f"for differences between zinnen only and {ch}:")
        # print(f"Processing {exp} for differences between zinnen and {ch}...")
        ##notice that in this comparison X is always ch_zinnen_mean
        x=
    dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{exp}_ch_only_zinnen"]
        ##notice that in n this comparison y is ch_woorden_mean or
    ch_intersection_mean
        y=dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{exp}_ch_{ch}"]

    diff=x-y
    # Permutation test
    res = permutation_test(
        (diff, np.zeros_like(diff)),
        #
        statistic=paired_statistic,
```

```

        vectorized=False,
        n_resamples=10000,
        alternative=alternative,
        random_state=42
    )

    # Cohen's d
    mean_diff = np.mean(diff)
    std_diff = np.std(diff, ddof=1)
    cohens_d = mean_diff / std_diff

    print(f"    Statistic value: {res.statistic:.6f}")
    print(f"    p-value          : {res.pvalue:.5f}")
    print(f"    Cohen's d          : {cohens_d:.3f}")

```

```

DEPENDENT analysis in {metric_name}
in zinnen condition
for differences between zinnen only and intersection:
    Statistic value: 0.041475
    p-value          : 1.00000
    Cohen's d        : 0.986
for differences between zinnen only and only_woorden:
    Statistic value: 0.066047
    p-value          : 1.00000
    Cohen's d        : 2.637
in woorden condition
for differences between zinnen only and intersection:
    Statistic value: 0.034636
    p-value          : 1.00000
    Cohen's d        : 0.888
for differences between zinnen only and only_woorden:
    Statistic value: 0.062151
    p-value          : 1.00000
    Cohen's d        : 2.564

```

2.1.2 1.1.2 If we assume Independency

```

[48]: ##if we assume independency
print(f"INDEPENDENT analysis in {metric_name}")
for exp in experimental_condition:
    print(f"in {exp} condition")
    for ch in type_channel:
        print(f"for differences between zinenn and {ch}:")
        # print(f"Processing {exp} for differences between zinenn and {ch}...")
        ##notice that in this comparison X is alwas ch_zinnnen_mean
        x=□
        dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{exp}_ch_only_zinnen"]

```

```

    ##notice that in n this comparison y is ch_woorden_mean or
    ↪ ch_intersection_mean
    y=dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{exp}_ch_{ch}"]

    # Hacemos la prueba de permutación
    res = permutation_test(
        (x, y),
        statistic=diff_means,
        vectorized=False,
        n_resamples=10000,
        alternative=alternative,
        random_state=42
    )

    # Calcular Cohen's d (independent)
    mean_x, mean_y = np.mean(x), np.mean(y)
    std_x, std_y = np.std(x, ddof=1), np.std(y, ddof=1)
    n_x, n_y = len(x), len(y)

    pooled_std = np.sqrt(((n_x - 1) * std_x**2 + (n_y - 1) * std_y**2) /
    ↪ (n_x + n_y - 2))

    if difference == "normal":
        cohens_d = (mean_x - mean_y) / pooled_std
    else:
        cohens_d = (mean_y - mean_x) / pooled_std

    # Imprimir resultados
    print(f"    Statistic value: {res.statistic:.6f}")
    print(f"    p-value           : {res.pvalue:.5f}")
    print(f"    Cohen's d           : {cohens_d:.3f}")

```

```

INDEPENDENT analysis in acw_0
in zinnen condition
for differences between zinenn and intersection:
    Statistic value: 0.041475
    p-value         : 0.94751
    Cohen's d       : -0.554
for differences between zinenn and only_woorden:
    Statistic value: 0.066047
    p-value         : 0.99770
    Cohen's d       : -1.021
in woorden condition
for differences between zinenn and intersection:
    Statistic value: 0.034636
    p-value         : 0.90311
    Cohen's d       : -0.447

```

```

for differences between zinenn and only_woorden:
    Statistic value: 0.062151
    p-value       : 0.99500
    Cohen's d     : -0.912

```

2.2 1.2 Using average in both conditions

2.2.1 1.2.1 If we assume dependency

```

[49]: print("DEPENDENT analysis in {metric_name}")
      for ch in type_channel:
          print(f"for differences between zinnen only and {ch}:")
          # print(f"Processing {exp} for differences between zinenn and {ch}...")
          ##notice that in this comparison X is always ch_zinnnen_mean

          # # #      x=
          ↪ dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{exp}_ch_only_zinnen"]
          # # # ##notice that in n this comparison y is ch_woorden_mean or
          ↪ ch_intersection_mean
          # # # y=dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{exp}_ch_{ch}"]
          x=
          ↪ (dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{experimental_condition[0]}_ch_only_
          ↪ 2
          y=
          ↪ (dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{experimental_condition[0]}_ch_{ch}
          ↪ 2

          diff=x-y

          # Permutation test
          res = permutation_test(
              (diff, np.zeros_like(diff)),
              #
              statistic=paired_statistic,
              vectorized=False,
              n_resamples=10000,
              alternative=alternative,
              random_state=42
          )
          # Cohen's d
          mean_diff = np.mean(diff)
          std_diff = np.std(diff, ddof=1)
          cohens_d = mean_diff / std_diff

          print(f"    Statistic value: {res.statistic:.6f}")
          print(f"    p-value       : {res.pvalue:.5f}")

```

```
print(f"    Cohen's d      : {cohens_d:.3f}")
```

```
DEPENDENT analysis in {metric_name}
for differences between zinnen only and intersection:
    Statistic value: 0.038056
    p-value       : 1.00000
    Cohen's d     : 0.941
for differences between zinnen only and only_woorden:
    Statistic value: 0.064099
    p-value       : 1.00000
    Cohen's d     : 2.642
```

2.2.2 1.2.2 If we asume independency

```
[50]: print("INdependent analysis in {metric_name}")

for ch in type_channel:
    print(f"for differences between zinnen only and {ch}:")
    # print(f"Processing {exp} for differences between zinenn and {ch}...")
    ##notice that in this comparison X is always ch_zinnnen_mean
    x=
    ↪(dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{experimental_condition[0]}_ch_only
    ↪2
    y=
    ↪(dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{experimental_condition[0]}_ch_{ch}
    ↪2

    # Permutation test
    res = permutation_test(
        (x, y),
        statistic=diff_means,
        vectorized=False,
        n_resamples=10000,
        alternative=alternative,
        random_state=42
    )

    # Calculat Cohen's d (independent)
    mean_x, mean_y = np.mean(x), np.mean(y)
    std_x, std_y = np.std(x, ddof=1), np.std(y, ddof=1)
    n_x, n_y = len(x), len(y)

    pooled_std = np.sqrt(((n_x - 1) * std_x**2 + (n_y - 1) * std_y**2) / (n_x +
    ↪n_y - 2))

    if difference == "normal":
```

```

        cohens_d = (mean_x - mean_y) / pooled_std
    else:
        cohens_d = (mean_y - mean_x) / pooled_std

    # Imprimir resultados
    print(f"    Statistic value: {res.statistic:.6f}")
    print(f"    p-value          : {res.pvalue:.5f}")
    print(f"    Cohen's d          : {cohens_d:.3f}")

```

```

INdependent analysis in {metric_name}
for differences between zinnen only and intersection:
    Statistic value: 0.038056
    p-value        : 0.92651
    Cohen's d      : -0.501
for differences between zinnen only and only_woorden:
    Statistic value: 0.064099
    p-value        : 0.99640
    Cohen's d      : -0.968

```

3 2. Comparison of each channel type in both conditions

Before we compared differences of word and sentence channels. Now we are going to compare the channels with themselves in different experimental conditions

```

[51]: ## statistical test

for ch in type_channel:
    print(f"for differences in {metric_name} ch {ch} between zinnen and word_
    ↪condition:")
    # print(f"Processing {exp} for differences between zinenn and {ch}...")

    #ch value in zinnen condition
    x=□
    ↪dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{experimental_condition[0]}_ch_{ch}"]
    y=□
    ↪dict_metric_condition_subj_epoch_mean_ch_type_mean[f"X_{experimental_condition[1]}_ch_{ch}"]

    print(f"X is: X_{experimental_condition[0]}_ch_{ch}", x.shape)
    print(f"Y is: X_{experimental_condition[1]}_ch_{ch}", y.shape)

    if difference=="normal":
        diff=x-y
    elif difference=="inverse":
        diff=y -x
        # Permutation test
    res = permutation_test(

```



```

        (diff, np.zeros_like(diff)),
        statistic=paired_statistic,
        vectorized=False,
        n_resamples=10000,
        alternative=alternative,
        random_state=42
    )

    # Calcular Cohen's d para muestras emparejadas
    mean_diff = np.mean(diff)
    std_diff = np.std(diff, ddof=1)

    if difference == "normal":
        cohens_d = (mean_x - mean_y) / pooled_std
    else:
        cohens_d = (mean_y - mean_x) / pooled_std

    # Imprimir resultados
    print(f"    Statistic value: {res.statistic:.6f}")
    print(f"    p-value          : {res.pvalue:.5f}")
    print(f"    Cohen's d          : {cohens_d:.3f}")

```

for differences in acw_0 ch intersection between zinnen and word condition:

X is: X_zinnen_ch_intersection (18,)

Y is: X_woorden_ch_intersection (18,)

Statistic value: 0.000996

p-value : 0.70003

Cohen's d : -0.968

for differences in acw_0 ch only_woorden between zinnen and word condition:

X is: X_zinnen_ch_only_woorden (18,)

Y is: X_woorden_ch_only_woorden (18,)

Statistic value: -0.001947

p-value : 0.19628

Cohen's d : -0.968